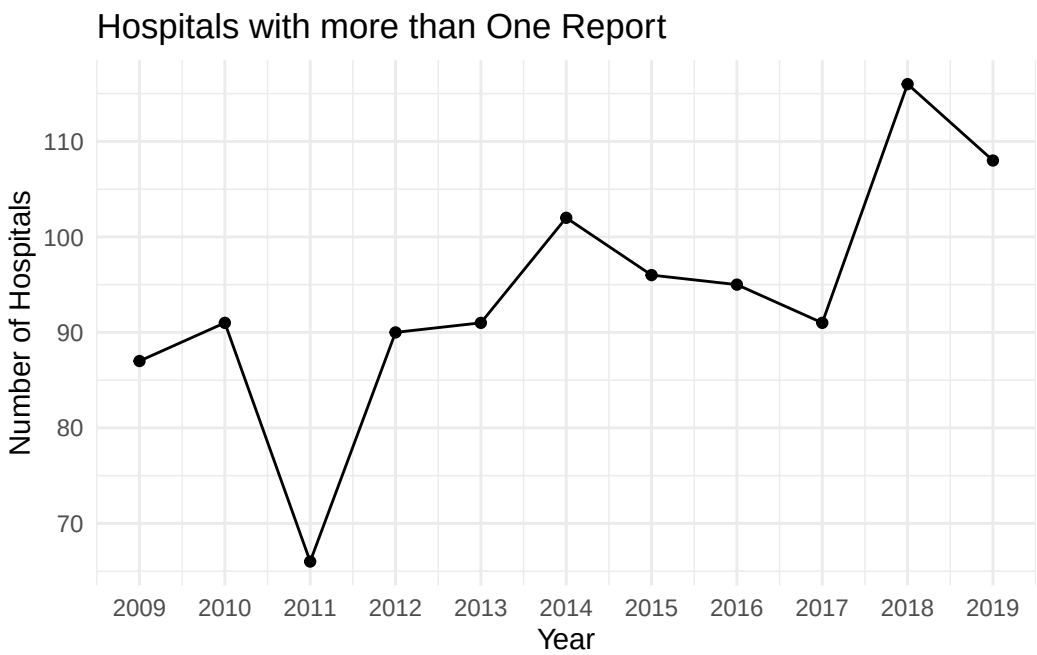
Homework 4 submission 3

Question 1

how many hospitals filed more than one report in the same year? Show answer as a line graph of the number of hospitals over time.



Question 2

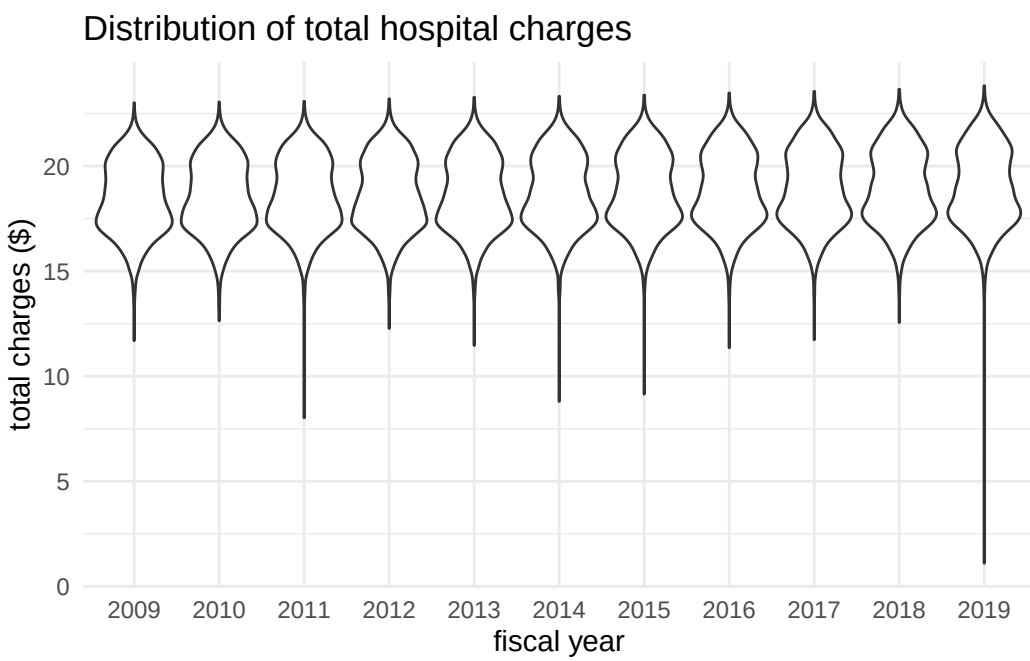
after removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

```
[1] "Total unique hospitas IDs 6897"
```

```
# A tibble: 11 x 2
    year hospitals
   <dbl>     <int>
 1  2009      6100
 2  2010      6103
 3  2011      6097
 4  2012      6140
 5  2013      6066
 6  2014      6064
 7  2015      6051
 8  2016      6091
 9  2017      6083
10  2018      6042
11  2019      6035
```

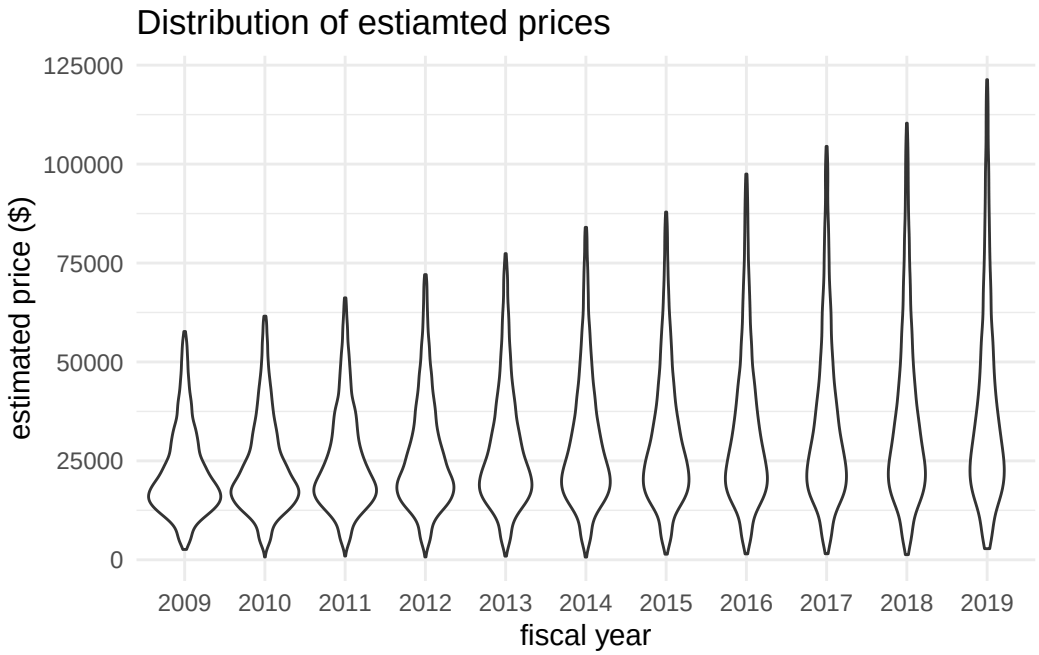
Question 3

What is the distribution of total charges (tot_charges) in each year? Show results in a violin plot with charges on the y-axis and years on the x-axis



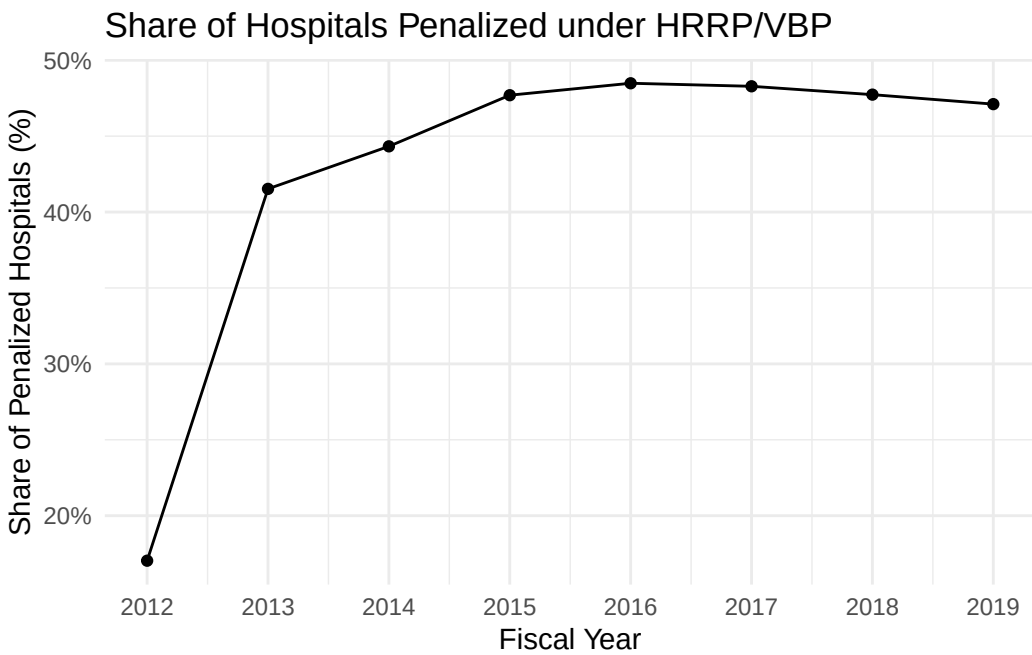
Question 4

what is the distribution of estimated prices in each year? Present results with a violin plot (be sure to do something about outliers and/or negative prices in the data)



Question 5

What share of hospitals penalized under HRRP/VBP? Provide a graph showing the share of penalized hospitals from 2012-2019.



Question 6

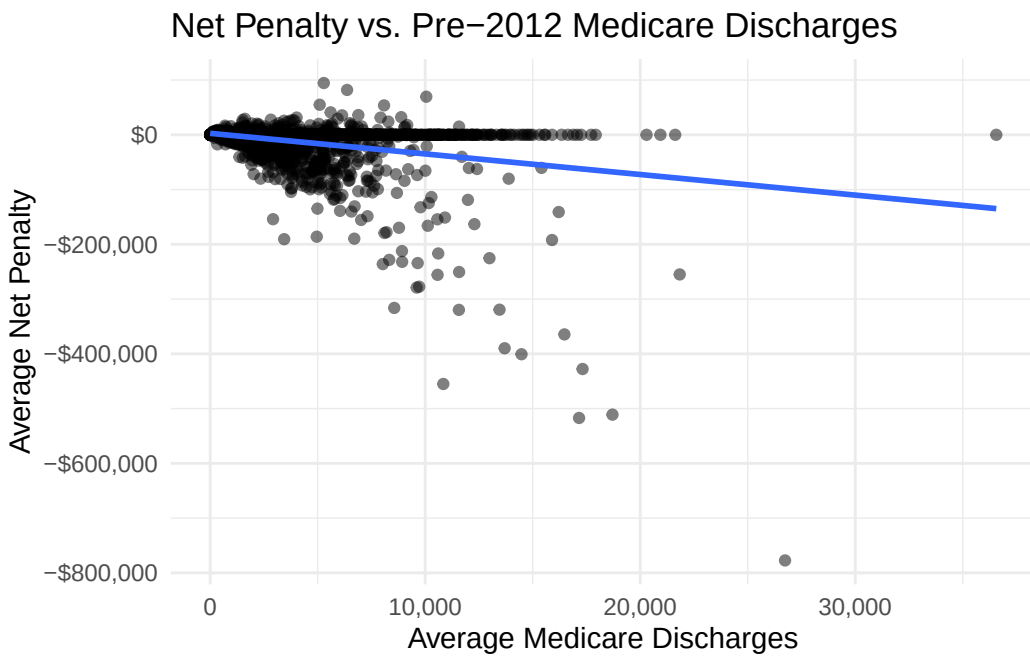
Provide a summary of OLS estimates of the effect of net penalties on price changes. Present your results in a table with three different specifications: 1) a "baseline" specification using only net penalty as a covariate; 2) "baseline" specification plus the pre-penalty (2009-2011) mean bed size; 3) "baseline" specification plus bed size plus pre-penalty (2009-2011) average Medicaid discharges.

	model1	model2	model3
	Baseline	with Bed Size	with Medicaid
Dependent Var.:	price_change	price_change	price_change
Constant	259.6 (523.4)	222.6 (529.4)	380.3 (467.8)
net_penalty	0.0058 (0.0131)	0.0055 (0.0131)	0.0023 (0.0095)
avg_beds		0.1554 (0.3313)	0.1099 (0.2398)
avg_mcaid_discharge			0.1462 (0.1746)
S.E. type	IID	IID	IID
Observations	2,519	2,519	2,508
R2	7.64e-5	0.00016	0.00044
Adj. R2	-0.00032	-0.00063	-0.00076

```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 7

Provide a scatterplot of net penalty against pre-2012 Medicare discharges.



Question 8

Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties. Present your results in a table with three different specifications as in Question 6.

```
                                     fs1                    fs2
                                Baseline          with Bed Size
Dependent Var.:                net_penalty            net_penalty

Constant             -5,314.5*** (1,104.4) -5,312.0*** (1,105.0)
avg_mcare_discharge      4.244*** (0.2313)     4.246*** (0.2326)
avg_beds                                        -0.0369 (0.4753)
avg_mcaid_discharge
------------------- --------------------- ---------------------
```

```
S.E. type                              IID                   IID
Observations                         2,519                 2,519
R2                                 0.11802               0.11802
Adj. R2                            0.11767               0.11732

                                       fs3
                               with Medicaid
Dependent Var.:                  net_penalty

Constant            -4,784.7*** (1,120.9)
avg_mcare_discharge     4.676*** (0.2671)
avg_beds                  0.0068 (0.4756)
avg_mcaid_discharge     -1.287** (0.3933)
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S.E. type                              IID
Observations                         2,508
R2                                 0.12161
Adj. R2                            0.12056
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


                                  rf1              rf2              rf3
                             Baseline    with Bed Size    with Medicaid
Dependent Var.:           price_change     price_change     price_change

Constant            -1,127.5 (772.9) -1,132.4 (773.3)   -325.9 (566.1)
avg_mcare_discharge 0.4003* (0.1619) 0.3964* (0.1628) 0.2990* (0.1349)
avg_beds                               0.0748 (0.3326)  0.0697 (0.2402)
avg_mcaid_discharge                                    -0.0642 (0.1986)
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S.E. type                         IID              IID              IID
Observations                    2,519            2,519            2,508
R2                            0.00242          0.00244          0.00237
Adj. R2                       0.00203          0.00165          0.00118
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 9

Provide a summary of IV estimates of the effect of net penalties on price changes. Again present your results in a table with the three different specifications as in Questions 6 and 8.

```
                                iv1              iv2              iv3
                           Baseline    with Bed Size         Medicaid
Dependent Var.:         price_change     price_change     price_change

Constant              -626.3 (640.2)   -636.5 (641.8)   -19.99 (504.4)
net_penalty         0.0943* (0.0385) 0.0934* (0.0387) 0.0639* (0.0291)
avg_beds                               0.0782 (0.3358)  0.0692 (0.2425)
avg_mcaid_discharge                                     0.0181 (0.1851)
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S.E. type                        IID              IID              IID
R2                          -0.01803         -0.01762         -0.01623
Observations                   2,519            2,519            2,508
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 10

Briefly explain the "Local" ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The local ATE refers to the effect of the penalties on price changes for the hospitals whose penalty status was changed by the pre-2012 medicare discharges. The ATE would be the impact of penalties on any hospital chosen randomly; the local ATE is weighted towards larger Medicare-dependent hospitals.